

Yash Agrawal

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EDUCATION

- **Hemvati Nandan Bahuguna Garhwal University, Srinagar** 2016 – 2019
Bachelor of Computer Applications

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript
- **Frontend:** React.js, Redux, Tailwind CSS, HTML5, CSS3, Bootstrap, Next.js, Figma, Three.js
- **Backend:** Node.js, Express.js, REST APIs, GraphQL
- **Databases:** MySQL, MongoDB
- **Tools & Platforms:** Git, GitHub, AWS

EXPERIENCE

- **SISO India Private Limited** June 2024 – Present
Software Engineer
 - Revamping a **tunnel construction monitoring platform** with advanced features including multi-language (i18n) support, dynamic theming, and interactive dashboards using React.js, Redux, Tailwind CSS, and TypeScript.
 - Developed and integrated **3D visualization modules using Three.js** for BIM-based project insights, enabling better tracking of construction progress and spatial data.
 - Implemented **map-based data visualization** using Maptiler APIs, building interactive geographic views and 3D globe integrations for real-time project monitoring.
 - Collaborating with a Switzerland-based team via GitHub, translating business requirements into scalable frontend solutions and aligning with backend APIs and architecture.
 - Developed and maintained backend APIs using Node.js and Express.js, supporting frontend modules and business workflows across the application.
 - Optimized application performance using **code splitting, lazy loading, memoization, and bundle optimization techniques**, significantly improving load time and user experience.
 - Designed and built **scalable, reusable component architecture** following best practices, improving maintainability and reducing development time across features.
 - Integrated **REST APIs and asynchronous data handling**, ensuring smooth data flow and state management across complex UI interactions.
 - Contributed to **UI/UX enhancements**, focusing on responsive design, accessibility, and consistent user experience across devices.
 - Worked on **4D/5D BIM concepts**, enabling visualization of construction timelines and progress within the application.
 - Leveraged AI tools like ChatGPT, Claud AI and Cursor to accelerate development, debug complex issues, and improve overall engineering productivity.
- **NIC (Akal Information & Velocis Systems Pvt. Ltd.)** July 2022 – May 2023
Software Developer
 - Developed and maintained single-page web applications using React.js, Redux, Tailwind CSS, HTML5, and JavaScript.
 - Implemented state management with Redux and Context API for complex application logic.
 - Integrated backend APIs to fetch, update, and manage data using RESTful and GraphQL endpoints.
 - Built an admin panel for analytics with interactive data tables and charts using React, Node.js, and MySQL.

- Optimized application performance using code splitting, lazy loading, and memoization techniques, ensuring smooth rendering and improved load times for large-scale applications.

- **Helloverify India Private Limited**

Jan 2022 – July 2022

- *Executive Data Analyst*

- Produced high-volume metrics, data, and detailed reports for client deliverables.
- Ensured 99% accuracy through rigorous data validation processes.
- Conducted data research and analysis using Power BI, MySQL, and Excel for reporting insights.
- Applied strong analytical skills to identify and resolve complex data challenges.
- Automated reporting workflows with Excel macros and SQL scripts, saving significant manual processing time.

PROJECTS

- **Portfolio Website (Link):** Built a responsive portfolio app using React.js and Tailwind CSS, deployed on Netlify to showcase work and skills.
- **3D BIM Model Viewer (Link):** Developed an advanced 3D BIM viewer using React.js, TypeScript, and Three.js to render and interact with complex construction models in real time. Implemented features such as model navigation (zoom, pan, rotate), object isolation, and element selection for enhanced usability. Built advanced tools including **sectioning, measurement, federated model handling, markup annotations, and view cube controls** to improve model exploration and analysis. Optimized performance for large-scale models using lazy loading and efficient rendering techniques. Designed a responsive and intuitive UI using Tailwind CSS with multi-language and theme support.
- **Query Management Portal (Link):** Built a query management system with authentication, authorization, JWT-based sessions, and data tables using React.js, Node.js, Express.js, and MySQL.
- **E-commerce Webpage:** Developed a MERN stack-based e-commerce prototype with product listing, authentication, and MongoDB for data storage.
- **Weather Pro Application:** Created a weather forecasting app using React.js and OpenWeather API to display real-time weather data.

CERTIFICATIONS

- Prompt Engineering for Developers (DeepLearning.AI)
- Advanced Frontend React.js (Ducat, Noida)
- Full Stack Web Development (Coding Blocks, CodeHelp)
- MERN Full Stack Web Development (CodeHelp)
- HTML, CSS, and JavaScript for Web Developers (Coursera)